

Character Controlling AI

AI Movement

Automatically causes AI to look at the LookTarget and move to MoveTarget. Requires Navmesh Agent.

AI has modified behavior for when it encounters an off mesh link. It moves forward and drops if the off mesh link end is close enough to be reached without jumping, and jumps if the distance from self to off mesh link end is too far to be made by dropping.

AI Action

Action determines where the agent will move to, where the agent will aim at, and what input it should use for the ability. The exact logic used is expected to be implemented manually.

DetermineAim

Abstract void method which sets where the aim target will be
Result: sets position of AimTarget

DetermineMovement

Abstract void method which sets where the move target will be
Result: sets position of MoveTarget

DetermineInput

Abstract void method which calls input to InputCall to cause AI to use active abilities.
Result: Set ability to use, ability input, and inputHoldType

The behavior of the agent can be set by setting the DetermineAim, DetermineMovement, and DetermineInput object in the AI Action.

ConditionCheck

Static class containing a method to check if a condition is true or false.

CheckIfConditionTrue

Use known context and a parameter (if applicable) to check if a given condition is true

Input: decisionCondition, float, KnownContext

Result: return boolean value true or false

DecisionTree

Tree which represents the decision making process that AI goes through in order to pick a AI Action to use.

Edit Decision Tree by making a new decision tree. Then open the tree editor through Window->AI->DecisionTreeEditor. Add new root node to begin

Nodes can be in either condition mode when Action is checked off, or be in action mode when Action is checked on. If in condition mode, it must have child nodes for the case where the condition is true or false. If on action mode, it must have an assigned all determine aim, movement, and input. If there is a point in the tree that does not end with an AI Action, MakeDecision method will return null.

MakeDecision

Decision tree starts from the root node and traverses through it based on if node's condition is true or false. Returns AI Action or Null as fallback

Input: n/a

Output: AI Action. Null as fallback

ObjectDetection

Produces a list of information based on what is nearby the character.

GetCurrentContext

Returns current context

Input: n/a

Result: current context

AIProcess

Continuously activates Object Detection, Decision Making, and AI Action to allow the character's AI to function.

PatrolLandmark

Set an object in the environment as a patrol landmark in order to make the character go near it when using Patrol movement action. The priority of the landmark can be set to make the agent prefer to be in one location over the other. This is useful for making the agent roll out to the objective from spawn. The agent uses the index of the priority set on the Char Stats, or 0 if not found for the landmark. Make sure the landmark has a collider so the agent can find it through its object detection.